

Methodological Note: A Linearized CGE Approach to a Real-Time Sectoral–Regional Layoff-Risk Dashboard

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1 Purpose

This note documents the methodology underlying a dashboard that maps current external and internal economic shocks into a sector $\times$ province matrix of employment change, used as a proxy for layoff risk. The design goal is a tool that updates in real time from live data feeds without requiring a full CGE re-solve on every refresh. This is achieved by pre-computing shock elasticities offline from IndoTERM and applying them linearly to observed shock magnitudes at run time.

2 Notation

- i Sector index, $i = 1, \dots, I$.
- r Region/province index, $r = 1, \dots, R$.
- j Shock index, $j = 1, \dots, J$ (e.g., coal export price, CPO export price, nickel export price, textile/garment/footwear export demand, import/input cost, minimum wage, fuel/electricity subsidy, macro/financial).
- $X_{j,t}$ Observed level of the raw driver underlying shock j at time t (e.g., the Newcastle coal price index in year t).
- x_j Percentage change in driver j , used as the dashboard’s live input.
- $\eta_{i,r,j}$ Calibrated elasticity: the percentage change in employment in sector i , region r , induced by a 1% change in shock j , holding all other shocks at zero.
- $e_{i,r,j}$ Realized employment effect of shock j on sector i , region r , at the current value of x_j .
- $E_{i,r}$ Total employment change in sector i , region r , summed across all active shocks. This is the dashboard’s primary output and the basis of the heatmap.

3 Step 1: Offline Calibration of Elasticities

For each shock j , IndoTERM is solved once with a 1% perturbation applied to shock j alone, all other shocks held at their base-case (zero) values. The resulting percentage change in sectoral-regional employment is recorded as the elasticity:

$$\eta_{i,r,j} = e_{i,r,j} \Big|_{x_j=1\%}, \quad \forall i, r, j. \quad (1)$$

This produces an $I \times R \times J$ elasticity tensor. This step is run offline (outside the dashboard) and only needs to be refreshed when the underlying CGE model, its closure, or its base-year SAM is updated — not on every dashboard refresh.

Regional decomposition shortcut (optional). For shocks whose regional incidence is primarily driven by each region’s sectoral employment composition rather than by genuine region-specific transmission (e.g., a national minimum-wage shock, as opposed to a nickel export price shock that hits producing provinces directly), the calibration burden can be reduced via

$$\eta_{i,r,j} \approx \eta_{i,j} \cdot s_{r,i}, \quad (2)$$

where $\eta_{i,j}$ is a national sector-level elasticity (no regional disaggregation needed in the CGE run) and $s_{r,i}$ is region r ’s share of national employment in sector i from the base-year SAM. This is a modeling choice to be made shock-by-shock, not applied uniformly, and should be flagged in the elasticity metadata so it is auditable later.

4 Step 2: Live Shock Magnitude Calculation

On each dashboard refresh, the system retrieves the latest available observation of each raw driver, $X_{j,t}$, and the corresponding value one year prior, $X_{j,t-1}$, and computes the percentage change:

$$x_j = 100 \times \frac{X_{j,t} - X_{j,t-1}}{X_{j,t-1}}. \quad (3)$$

Note on Equation (3). The correct percentage-change formula subtracts 1 (equivalently, the prior-period level) from the ratio before scaling by 100. A formula of the form $x_j = 100 \cdot X_{j,t} / X_{j,t-1}$ without the subtraction does *not* return zero when $X_{j,t} = X_{j,t-1}$, and will silently distort every downstream calculation. This should be enforced with an explicit unit test in the data pipeline: a synthetic series with no period-over-period change must return $x_j = 0$ exactly.

Frequency and alignment. t and $t - 1$ need not be calendar years if higher-frequency data are available (e.g., month-over-month or quarter-over-quarter), provided the elasticities in Step 1 are calibrated consistently with the chosen frequency, or are treated as approximately frequency-invariant for small, linearized shocks. Each shock series should carry an explicit timestamp and frequency tag so that mismatched vintages across j are visible to the user rather than silently blended.

5 Step 3: Real-Time Aggregation

The realized employment effect of each shock on each sector-region cell is the calibrated elasticity scaled by the observed shock magnitude:

$$e_{i,r,j} = \eta_{i,r,j} \cdot x_j. \quad (4)$$

The total employment change for sector i , region r , summed across all shocks currently active in the dashboard, is

$$E_{i,r} = \sum_{j=1}^J e_{i,r,j} = \sum_{j=1}^J \eta_{i,r,j} \cdot x_j. \quad (5)$$

$E_{i,r}$ is the matrix rendered as the heatmap: rows (or columns) indexed by sector i , the orthogonal axis indexed by province r , color intensity proportional to the magnitude (and sign) of employment change, used as the layoff-risk signal.

6 Computational Implication

The architecture deliberately separates a slow, infrequent step (Step 1: CGE solves, run offline, computationally expensive) from a fast, frequent step (Steps 2–3: a tensor contraction over pre-stored elasticities and live data, computationally trivial). This is what allows “press update” to return a result in real time: the dashboard never re-solves the CGE. The validity of this separation rests entirely on the linearity and additivity assumptions below, which is the price paid for real-time responsiveness.

7 Assumptions and Known Limitations

This additive-linearization design is adopted deliberately for a first-version, real-time dashboard, with the following assumptions made explicit so they can be revisited in later iterations.

1. **Local linearity.** Equation (5) assumes the employment response to each shock is linear in the shock’s magnitude, with a constant slope $\eta_{i,r,j}$ equal to the response observed at a 1% perturbation. This is exact only in the limit of infinitesimally small shocks. CGE responses are generally nonlinear once a closure rule binds differently at different shock magnitudes (e.g., the dual labor market closure — rigid-wage/ quantity-adjusting in formal manufacturing, underemployment-absorbing in the informal sector — may not respond proportionally as shock size increases). Historical shocks of interest (commodity price swings in coal, CPO, nickel) have often been large in percentage terms, well outside the neighborhood in which a 1%-calibrated slope is guaranteed to extrapolate cleanly.
2. **Additive separability across shocks.** Equation (5) sums effects across shocks as if they do not interact. In a general equilibrium setting with jointly clearing factor markets, this holds only approximately, and the approximation error grows when multiple shocks load onto the same factor or sector simultaneously (e.g., concurrent export demand shocks to textiles, garments, and footwear, all labor-intensive and regionally concentrated).
3. **Time-invariant elasticities.** $\eta_{i,r,j}$ is calibrated once from the current model structure, closures, and base-year SAM, and held fixed between recalibrations. Structural change in the economy (sectoral composition shifts, policy regime changes) will erode the accuracy of a stale elasticity tensor. A recalibration cadence should be defined rather than left implicit.
4. **Heterogeneous epistemic status of x_j inputs.** Not all shock variables enter through a CGE-calibrated elasticity in the same way. For example, PMI (Purchasing Managers’ Index) is not a CGE primitive and is treated as a reduced-form input via a separately estimated regression translation from its new-orders sub-index, rather than a CGE-simulated shock. Such inputs should be flagged distinctly in the elasticity metadata so a user does not implicitly treat a regression-derived coefficient with the same confidence as a simulated structural elasticity.
5. **No re-solve benchmarking (yet).** There is currently no periodic check of the additive linear approximation against a full multi-shock CGE re-solve. Before the dashboard is used

for any high-stakes policy communication, it is advisable to benchmark $E_{i,r}$ from Equation (5) against a true joint re-solve under at least a few realistic historical shock combinations, to establish an empirical error bound on the linear-additive approximation.

8 Upgrade Path (Future Versions)

- **Two-point calibration.** Run each shock at two magnitudes (e.g., 1% and 10%) and compare $\eta_{i,r,j}$ across the two. If stable, linear extrapolation is empirically supported for that shock; if not, fit a quadratic correction $e_{i,r,j} = \eta_{i,r,j}x_j + \frac{1}{2}\gamma_{i,r,j}x_j^2$ using the two calibration points.
- **Interaction terms.** For shock pairs known to co-move or to load on the same factors/sectors, estimate a cross term $\eta_{i,r,j,k}$ from a joint CGE perturbation, and add $\sum_{j < k} \eta_{i,r,j,k} x_j x_k$ to Equation (5).
- **Periodic full re-solve benchmarking.** Establish a routine (e.g., quarterly) full CGE re-solve under realized shock combinations, and report the linear-additive approximation error to dashboard maintainers.
- **Elasticity versioning.** Track calibration date, IndoTERM model version, and base-year SAM version alongside each $\eta_{i,r,j}$, so dashboard outputs are reproducible and auditable against the model vintage used.

9 Summary

Step	Frequency	Computational cost
Calibrate $\eta_{i,r,j}$ via IndoTERM	Offline; on model/SAM update	High
Compute x_j from live data	Every dashboard refresh	Negligible
Compute $E_{i,r} = \sum_j \eta_{i,r,j}x_j$	Every dashboard refresh	Negligible
Render heatmap of $E_{i,r}$	Every dashboard refresh	Negligible

Table 1: Division of labor between offline CGE calibration and real-time dashboard computation.